

DevOnboard

Short Description

DevOnboard is a developer assisting platform that streamlines repository setup, documentation, environment validation, and onboarding tasks into one guided workflow.

Long Description

Onboarding a new engineer into an existing software project is expensive, time-consuming, and often inefficient. New developers must understand unfamiliar codebases, configure local environments, navigate outdated or scattered documentation, and learn project-specific workflows, while senior engineers spend valuable time answering questions and troubleshooting setup issues. As engineering teams grow, these repeated onboarding costs can become a significant drain on productivity and development capacity.

DevOnboard addresses this problem by transforming developer onboarding into a centralized, guided experience. Instead of relying on senior engineers to manually walk every new developer through a project, DevOnboard provides an interactive onboarding workflow that helps engineers understand the codebase, complete setup requirements, validate their development environment, navigate documentation, and track their progress independently.

The platform combines guided onboarding checklists, automated environment validation, commit history analysis, documentation-based Q&A, and progress tracking into a single workspace. Managers can also customize onboarding workflows and monitor completion, giving teams greater visibility into where new engineers are progressing or getting blocked.

Built with a modern React, TypeScript, Node.js, Express, and MongoDB stack, DevOnboard demonstrates a scalable approach to reducing the human effort required to onboard developers into complex software projects. By shifting onboarding from a largely manual process toward an automated workflow, the platform aims to help engineering teams reduce repetitive support work, shorten the path to developer

productivity, and scale their teams without scaling onboarding overhead at the same rate.

How IBM Bob was used

IBM Bob was used as an AI development agent throughout the design, planning, implementation, and validation of DevOnboard. We used Bob directly within the project repository to analyze the existing codebase, identify its architecture and project-specific conventions, and generate developer guidance through files such as AGENTS.md and Bob-specific rules. Bob examined the existing boilerplates and design system before creating a structured migration plan for converting the prototype into a React application.

During implementation, IBM Bob was used to scaffold the Vite + React + TypeScript + Tailwind CSS application, configure dependencies, establish the routing architecture, and create reusable UI components. Bob then built the application's planned screens, including the Welcome page, Setup Checklist, Commit History, Documentation Q&A, Environment Validator, Manager Settings, and Onboarding Completion workflows.

We also used IBM Bob as an iterative engineering assistant rather than only for code generation. When implementation issues occurred, Bob inspected the generated project, identified configuration and TypeScript problems, removed unnecessary template files, corrected imports and type declarations, and ran type-checking and production builds to validate the application.

This made IBM Bob a core part of our development workflow: it helped us move from an initial static prototype to a structured React application while continuously analyzing the repository, planning technical changes, generating implementation code, debugging issues, and validating the resulting system.